UNILAK RWAMAGANA CAMPUS

SOFTWARE ENGINEERING AND USER CENTRED DESIGN

Citizen Issue Reporting and Escalation System

CIRES

Milestone 2

System Requirements Report

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SECTION A SPECIFICATION CONTEXT

A1 Purpose and relationship to discovery

This specification defines the proposed first release of the Citizen Issue Reporting and Escalation System. CIRES will connect a resident's road or drainage report to officer assignment, visible progress, supervisory follow-up and an explained resolution. The requirements build on the reporting risks explored in Milestone 1. Those risks were developed from the topic proposal and fictional walkthroughs, and remain subject to review with actual users.

The document retains the identifiers and priorities in the supplied Milestone 2 presentation. It expands the meaning of each requirement and states how it will be checked. The use-case diagram summarises the participating roles, while the user stories connect the features to observable outcomes. The report also preserves the source's pending status for user validation and live backlog evidence.

A2 Scope of the first release

The pilot will operate in one configured area and accept two categories, Road and Drainage. Its users are residents, assigned officers and a supervisor. The included workflow begins when a resident submits a report and ends when the assigned officer records a resolution with an action summary. Each report can receive one overdue escalation to the supervisor. The pilot includes a history linking the case to its reference, assignment, progress and outcome.

Emergency dispatch, other issue categories, district-wide rollout, external agency integration, SMS alerts and multi-level escalation are outside this release. Queue filtering is a Should feature, while a weekly closure-count export is a Could feature. The scope therefore reserves the first implementation effort for the complete report-to-resolution journey.

A3 Operational assumptions and terms

A report is the stored account of a road or drainage issue. Its reference is the unique identifier used for later retrieval. The report owner is the resident who submitted it. The assigned officer is the staff member responsible for its progress. The due timestamp records when action is expected. Escalation is the single recorded flag that makes an overdue open case visible to the supervisor.

The workflow uses the states Received, Assigned, In progress and Resolved. Assignment places responsibility with an officer and includes a future due timestamp. Starting work changes Assigned to In progress. Resolution requires an action summary. The overdue flag does not replace these states. It identifies an Assigned or In progress case whose deadline has passed, while the work status continues to describe its operational stage.

For the pilot, accounts, role assignments and the service area are assumed to be configured before use. The supplied slides do not define a public registration process or a separate administrator interface. These activities are therefore treated as setup assumptions. The project also requires an agreed local reporting responsibility and a configurable overdue-check schedule before operational use.

SECTION B USER RESPONSIBILITIES AND WORKFLOW

The resident enters a category, location and description and receives a reference. They can retrieve their own report and read its current status and public updates. They do not assign officers or alter staff decisions. The assigned officer reads the case information, records that work has started and resolves an In progress report with an explanation of the action taken.

The supervisor assigns Received reports, sets future deadlines and reviews overdue open cases. Their queue provides the report context needed for follow-up. They may also use category and status filters if the Should feature is implemented. Weekly summary exports, when included, contain counts by category rather than individual resident details.

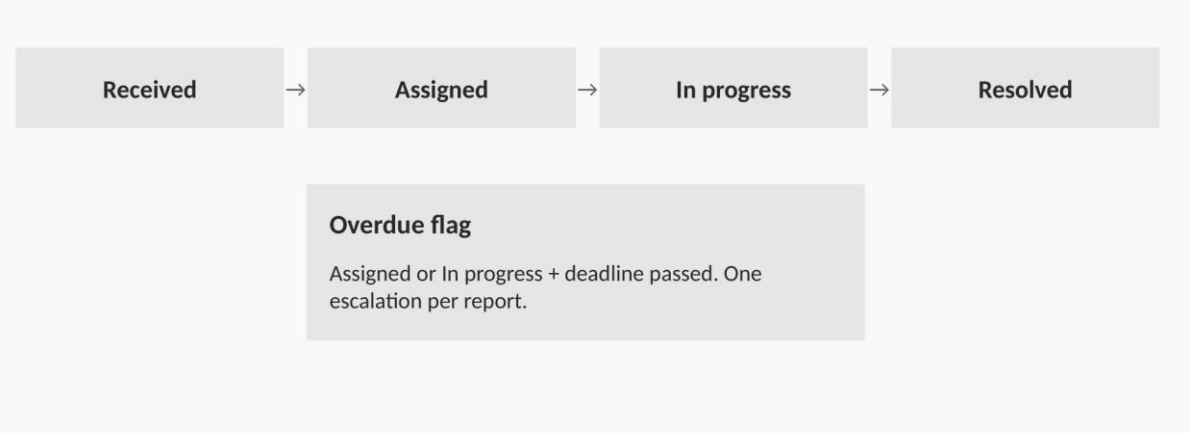


Figure 1 CIRES report workflow reproduced in monochrome

The sequence begins with a Received case and progresses through assignment and work to resolution. An overdue check examines the due time and the current state. It records an escalation only when an eligible open case has passed its deadline and has no prior escalation. A repeated check preserves the first escalation. Once resolution is recorded, the case is no longer eligible for the overdue open queue.

SECTION C FUNCTIONAL REQUIREMENTS

The functional requirements describe the behaviour that the pilot must provide. FR-01 to FR-06 are Must requirements. FR-07 is Should and FR-08 is Could. Each requirement is linked to one of the five reporting concerns, namely receipt, ownership, visibility, delay and closure.

FR-01 Receipt

Priority Must. The system shall accept a report with category, location and description and return a unique reference.

A signed-in resident will select Road or Drainage and supply a location within the pilot area. Successful validation creates one Received report. The reference gives the resident a stable identifier for progress enquiries. Missing fields and an unsupported location must produce errors without creating a case.

FR-02 Ownership

Priority Must. The system shall let the supervisor assign a Received report to a pilot officer with a future deadline.

The assignment will connect a valid officer account to the report and make it visible in the officer's queue. The report becomes Assigned. An invalid officer, a past deadline or an unauthorised request must not change the record. This requirement establishes the responsibility and timing on which later overdue checks depend.

FR-03 Visibility

Priority Must. The system shall show the owner the current status and dated public updates for their report.

The reference identifies the case, while the authenticated account determines whether the person may view it. The owner will see the latest state and public history without receiving private contacts or internal notes. This requirement addresses repeated progress enquiries by making the stored position available to the resident.

FR-04 Ownership

Priority Must. The system shall let the assigned officer change an Assigned report to In progress.

Only the current assigned officer can perform the change. The system records the new state and timestamp in the history. An officer assigned to another case must be unable to update it. The transition separates allocation of responsibility from confirmation that work has begun.

FR-05 Delay

Priority Must. The system shall flag each overdue Assigned or In progress report once and show it in the supervisor's queue.

Eligibility depends on the due timestamp, an open state and the absence of an existing escalation. Repeated checks must preserve one escalation record and its original timestamp. A Resolved case is excluded. The flag makes the overdue case visible for human follow-up rather than automatically resolving the underlying issue.

FR-06 Closure

Priority Must. The system shall require a public action summary before the assigned officer resolves an In progress report.

The summary must contain a meaningful nonblank explanation of action taken. Saving the resolution changes the state to Resolved and makes the summary available to the owner. A blank summary or an unauthorised officer must not close the case. The history remains available after resolution.

FR-07 Ownership

Priority Should. The system shall filter the supervisor's report list by category and status.

Both filters must apply together so that a user can identify, for example, Drainage cases that remain In progress. Clearing the filters restores the permitted list. This feature improves queue navigation after the core workflow is working.

FR-08 Closure

Priority Could. The system shall export weekly counts of resolved reports by category without resident identifiers.

The supervisor selects the week and receives a CSV containing the corresponding category counts. Only reports resolved within that reporting period contribute to the totals. The export excludes individual names, private contacts, internal notes and case descriptions. The reporting calendar and time zone must be fixed before implementation testing.

SECTION D QUALITY AND ACCEPTANCE TARGETS

The non-functional requirements define measurable conditions for the pilot. They are proposed targets rather than test results. The verification record must state the test environment, accounts, data and observation period so that the resulting evidence can be interpreted.

NFR-01 Access

All core actions require sign-in. Cross-account reads and unauthorised updates return no case data and make no changes.

Tests will use an owner, a different resident, an assigned officer, another officer and a supervisor. They must cover direct requests as well as normal navigation. Each private operation must check the account's relationship to the report. This is consistent with OWASP guidance that permission is evaluated on every request (OWASP Foundation, n.d.).

NFR-02 Privacy

Resident views and exports expose zero private contact fields or internal staff notes.

The check will inspect rendered pages, downloaded CSV files and returned response data. A field must not be treated as protected merely because the interface hides it. The owner's visible history must contain only material intended for the resident, while weekly exports retain the defined category counts.

NFR-03 Speed

At least 95% of report detail requests finish within 2 seconds in 50 trials with 1,000 stored cases.

The test will seed a dataset containing the supported categories and workflow states. At least 48 of 50 requests must meet the threshold. The report will identify the network and server conditions used. This measures detail retrieval and does not promise the same duration for all operations or deployments.

NFR-04 Accessibility

Core forms support keyboard operation, visible labels and 4.5:1 contrast for normal text.

The review will examine submission, assignment and update forms using keyboard navigation and contrast measurements. These criteria draw on selected WCAG 2.2 provisions for keyboard access and normal-text contrast (World Wide Web

Consortium, 2024). The pilot will record the selected checks without claiming a full accessibility certification.

NFR-05 Usability

Two first-time resident testers can each submit a valid report within 2 minutes without assistance.

The test begins when each participant opens the reporting form and ends when the reference appears. Both participants must complete the task within the limit without coaching. Their difficulties and outcomes must be documented. The source material provides no completed results for this test.

NFR-06 Availability

At least 99% of scheduled five-minute checks succeed during a seven-day pilot observation window.

Checks will run at a fixed five-minute interval against a defined pilot endpoint. The number of successful checks will be divided by the total scheduled checks for the observation period. A successful check must meet the documented response condition. Missing and failed observations must be reported so that the percentage is reproducible.

SECTION E PRODUCT BACKLOG

The user stories retain the priorities and effort estimates in the source presentation. Story points express relative team effort. They are planning estimates and are not measurements of hours. Each acceptance condition follows the Given, When and Then sequence so that the intended result can later become a test.

US-01 — Must — 3 pts

As a resident, I want to submit a road or drainage report so I can receive proof that it was received.

Given a signed-in resident enters valid details when they submit then one Received report is saved and its reference appears..

Traceability PP1 to FR-01.

A missing location, unsupported category or location outside the pilot boundary must produce a clear error and save no report.

US-02 — Must — 3 pts

As a supervisor, I want to assign a report with a due date so I can make ownership clear.

Given an unassigned Received report exists when the supervisor selects an officer and future deadline then the officer sees it in their queue and the report becomes Assigned..

Traceability PP2 to FR-02.

An invalid officer or a due timestamp that is not in the future must leave the existing Received report unchanged.

US-03 — Must — 2 pts

As a resident, I want to check my report so I can see its progress without calling.

Given a resident owns an assigned report when they open its reference then they see its current status and public history..

Traceability PP3 to FR-03.

An unrelated resident using the same reference must not receive the case details. Private fields must remain outside the owner view.

US-04 — Must — 2 pts

As an assigned officer, I want to start work on my case so I can show that action is underway.

Given a report is assigned to the officer when they mark it In progress then the new status and timestamp appear in its history..

Traceability PP2 to FR-04.

The status change must be refused when the requesting officer is not assigned to the report or when its state is no longer Assigned.

US-05 — Must — 3 pts

As a supervisor, I want to see overdue cases in one queue so I can follow up before they are forgotten.

Given an open assigned report has passed its deadline when the overdue check runs twice then exactly one escalation is recorded and visible to the supervisor..

Traceability PP4 to FR-05.

A repeated or concurrent overdue check must not add another escalation. Resolved reports must not remain in the overdue open queue.

US-06 — Must — 3 pts

As an assigned officer, I want to close a case with an action summary so I can explain the outcome to the resident.

Given an In progress report belongs to the officer when they save a nonblank action summary and resolve it then the report becomes Resolved and the resident can read the summary..

Traceability PP5 to FR-06.

The application must reject a blank action summary. The resolution and its history entry must remain consistent if an attempt is repeated.

US-07 — Should — 5 pts

As a supervisor, I want to filter the officer queue by category and status so I can find cases for follow-up.

Given the queue contains mixed categories and statuses when the supervisor applies both filters then only matching reports are shown..

Traceability PP2 to FR-07.

The filter test must include reports that match only one selected condition, confirming that both category and status are applied.

US-08 — Could — 5 pts

As a supervisor, I want to export a weekly closure summary so I can review the pilot's workload.

Given resolved cases exist within and outside the selected week when the supervisor exports that week then the CSV contains only counts by category for that week..

Traceability PP5 to FR-08.

The export test must include cases resolved before, within and after the selected week. The file must contain category totals for the selected period only.

SECTION F RELEASE PLANNING

The source backlog totals 26 estimated points. The six Must stories account for 16 points, approximately 62% of planned effort. Session 4 covers submission, assignment and resident tracking, totalling 8 points. Session 5 covers officer progress, overdue escalation and resolution, also totalling 8 points. This order establishes the records and ownership before implementing the later state changes.

US-07 accounts for 5 Should points and supports category and status filtering after the core workflow is complete. US-08 accounts for 5 Could points and provides the weekly closure export. Emergency dispatch, SMS alerts and multi-level escalation remain outside the release. These priorities keep implementation focused on a complete resident-to-resolution journey.

SECTION G USE CASE ANALYSIS

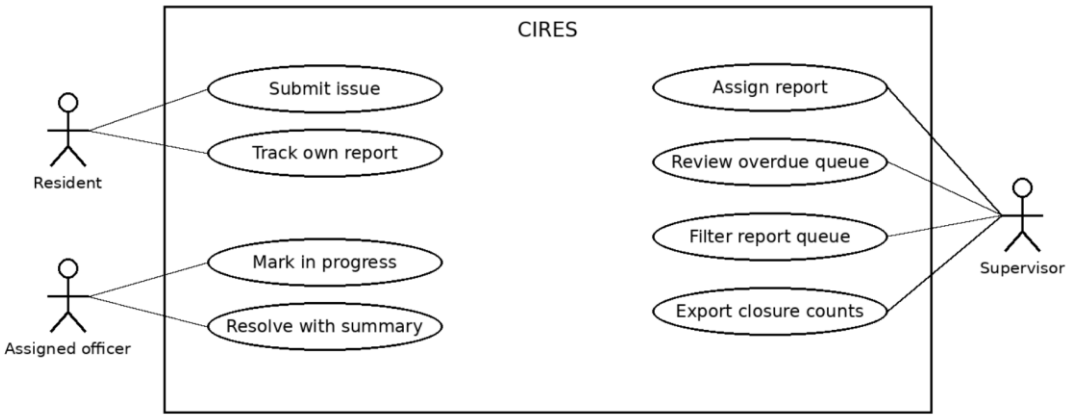


Figure 2 CIRES use-case diagram retained from the supplied presentation

The use-case diagram places the resident, assigned officer and supervisor outside the system boundary. The resident submits and tracks a report. The officer starts work and resolves a case with a summary. The supervisor assigns reports, reviews overdue cases and may use queue filtering and closure exports. Association lines identify participation. The overdue check is an internal system behaviour supporting the supervisor's review goal.

UC-01 Submit an issue

The primary actor is the resident and the related requirement is FR-01. The preconditions are that the resident is signed in and that the location is within the pilot area. The trigger is the decision to report a road or drainage concern.

In the main flow, the resident chooses Road or Drainage, enters the location and description and submits the form. CIRES validates the required information and saves a Received record. It then displays a unique reference. The successful postcondition is a stored report linked to the resident and available for supervisory assignment.

In the alternative flow, a required field is missing or the location is unsupported. The system displays field errors and saves no report. The resident can correct the details and submit again. A reporting reference confirms receipt, while the later assignment and resolution remain separate steps.

UC-02 Review an overdue case

The primary actor is the supervisor. The related requirements are FR-02, FR-05 and FR-06. The report must have an assigned officer and a due timestamp. Its deadline must have passed while its state remains Assigned or In progress.

In the main flow, the scheduled check records one escalation and shows the report in the overdue queue. The supervisor opens the report and reviews its assigned officer, due time and history. Follow-up takes place through the agreed operating process. The case remains visible while open and is removed from the overdue queue after an authorised resolution is recorded.

In the alternative flow, a report is already Resolved when checked, so it is skipped. If an escalation already exists, repeated checks preserve its timestamp. A check that overlaps with resolution must leave the report closed and outside the overdue open queue. The escalation remains part of the report history for later reference.

SECTION H TRACEABILITY AND TEST COVERAGE

PP1 concerns receipt and connects FR-01 to US-01 and T01. Its test verifies that valid details create one Received report and a unique reference. PP2 concerns ownership and connects FR-02, FR-04 and FR-07 to US-02, US-04 and US-07. T02 checks assignment, T04 checks the authorised progress transition and T07 checks combined filtering when the Should feature is implemented.

PP3 concerns visibility and connects FR-03 to US-03 and T03. The owner must see the current state and public history while an unrelated account receives no case details. PP4 concerns delay and connects FR-05 to US-05 and T05. Repeated overdue checks must produce one escalation record.

PP5 concerns closure and connects FR-06 and FR-08 to US-06 and US-08. T06 verifies that a nonblank summary is required before resolution and is visible to the resident afterwards. T08 checks the weekly category counts if the Could feature is implemented. The traceability links allow a requirement change to be followed through its story, acceptance condition and test.

SECTION I USER REVIEW AND COMPLETION STATUS

The first validation walkthrough is intended for a resident who can discuss reporting a blocked drain and finding its status. The participant should review the Must list, explain what confirms receipt and identify the update that would show that action had been taken. The resulting notes must record the participant code, task, difficulty, requested change and affected requirement.

The second walkthrough is intended for an assigned officer. It should examine an assigned road issue approaching its deadline and ask what information is needed to decide the next action. This review must test whether location, original description, deadline and history are sufficient for the proposed process. The supervisor's escalation workflow also needs confirmation within the operating arrangement.

The supplied presentation states that user feedback is pending. The report therefore does not invent review dates, participant reactions or requested changes. The live Jira or Trello board link and screenshot also remain to be added. The story specifications and effort estimates provide a basis for that board but are separate from evidence that a live board has been created.

After the actual reviews and backlog evidence are recorded, the team should retain this report with the editable diagram and exported image in the milestone folder. The m2 GitHub release will identify the submitted version. Readiness for the design stage will depend on whether users understand the Must workflow and whether the team can explain the response to each requested change.

References

Mugiraneza, Y. H. (2026). SECD session introductions [PowerPoint slides]. UNILAK.

OWASP Foundation. (n.d.). Authorization cheat sheet. OWASP Cheat Sheet Series.

Student project guide — Software engineering and user-centred design. (2026). [Course guide]. UNILAK.

World Wide Web Consortium. (2024). Web content accessibility guidelines (WCAG) 2.2. W3C Recommendation.